Detail-Optimized Super-Resolution Reconstruction-Based Multistage Training Strategy for Remote Sensing Semantic Segmentation

Baikai Sui, Yungang Cao, *Member, IEEE*, Haibo Cheng, Shuang Zhang, Jun Zhu, and Yakun Xie

***Abstract*— Low resolution is a major factor that negatively impacts the accuracy of remote sensing (RS) interpretation. High-quality super-resolution reconstruction (SRR) can help alleviate this problem. In this study, we present a multistage semantic segmentation training strategy called SRSSTS, which is based on detail-optimized SRR. SRSSTS addresses the challenges of poor reconstruction quality and low semantic segmentation (SS) accuracy of RS images. Our approach involves constructing a generative adversarial network (GAN)-based perceptual-loss-dominated SRR network, which generates texture-realistic, high-quality high-resolution images from low-resolution RS images. We then input the generated images into an SS network in steps and propose a three-stage joint loss to generate segmentation results at different resolutions. SRSSTS is straightforward and efficient, and it can be applied to any SS backbone networks. Additionally, this strategy does not increase computational resources compared to other SR-based segmentation methods, since the SRR network is trained separately. We evaluated the effectiveness of our designed SRR network and SRSSTS on five RS datasets (ISPRS Potsdam and Vaihingen, WHDLD, LoveDA Rural and Urban) with different resolutions, achieving excellent performance compared to an SS model for a single task and other SR-based segmentation methods. Moreover, we observed that the learned perceptual image patch similarity (LPIPS) of the super-resolution (SR) reconstructed images, i.e., the stronger the perception ability, the more helpful it is for the SS task.**

***Index Terms*— remote sensing (RS), semantic segmentation (SS), super resolution (SR), training strategy (TS).**

I. Introduction

With the rapid development of remote sensing (RS) technology, the volume of optical RS image data has proliferated, and RS interpretation has played an important role in image processing and analysis. And in recent years, with the rapid development of deep learning (DL), RS interpretation is gradually intelligent, and the mainstream interpretation methods of semantic segmentation (SS) and target detection based on convolutional neural networks have been formed.

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The authors are with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu 611756, China (e-mail: 13012482890@163.com; yungang@swjtu.edu.cn; haibocheng2000@163.com; Zhang_Shuang1999@163.com; zhujun@swjtu.edu.cn; yakunxie@my.swjtu.edu.cn).

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The task of segmenting RS images is highly challenging due to their low resolution, unique characteristics, such as the diverse types of features, complex spatial relationships, spectral sequences, and the impact of external factors such as cloud cover and sensor quality. Traditional DL models and training strategies may not be sufficient to achieve fine-grained segmentation of multiresolution RS images, and as a result, researchers have invested a significant amount of effort into addressing this issue. Such as, coarse-to-fine learning strategy. [1], [2], refine the segmentation results from the whole to the local; local and global attention perception [3], [4], saving computational effort and facilitating training are especially important for RS images with high pixel level; multiscale feature fusion [5], [6], the images are sampled at different sizes, with large objects segmented at low resolution and small objects segmented at the underlying high resolution; edge reinforcement learning, optimize RS image segmentation boundary. There are generally three ways: one is multinetwork multibranch joint training [7], the second is edge enhancement followed by segmentation [8], and the third is segmentation followed by edge adjustment [9]; data augmentation [10], improving data diversity makes it easier to train complex RS data. However, the own resolution of RS image is an important factor limiting its interpretation accuracy, and as a solution, it is especially critical to obtain the corresponding high-resolution RS image. Super-resolution reconstruction (SRR) is the key technology to solve this problem.

With the development and maturity of DL-based SRR technology, the loss of low-resolution to high-resolution conversion is gradually reduced. In 2014, the proposal of SRCNN [11] based on convolutional neural network made a breakthrough in SRR, and then improved versions based on this network have emerged, like FSRCNN [12], ESPCN [13], VDSR [14], LapSRN [15], EDSR [16], etc., which also made the PSNR-oriented super-resolution (SR) field become hot. After 2017, attention has been paid to SRR using generative adversarial networks (GANs), from SRGAN [17] to ESRGAN [18] to Real-ESRGAN [19], with relatively mature super-segmentation algorithms that generate high-resolution images with more detailed information and finer picture quality, bringing important applications in areas such as HDTV, medical imaging, and RS imaging. As far as RS interpretation is concerned, the images after SR can be better recognized and detected, and most researches tend to SR first and then recognize or double branch for large and small objects' recognition

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[20], [21], [22]. Instead, we focus on how to integrate SRR into RS SS work. In fact, this idea is similar to multiscale feature fusion learning, in which images at multiple scales and multiple resolutions are learned and feature extracted separately. Today's multiscale learning approaches typically involve downsampling the input image to obtain multiple low-resolution features at different scales. However, it is widely believed that obtaining high-resolution image features is more advantageous for tasks such as SS, especially for recognizing small targets. Unfortunately, current upsampling techniques, such as conventional interpolation resizing [23], [24] or upsampling convolutional layers, are often unable to produce high-quality images that meet the requirements of these tasks. As a result, there is a need for new approaches that can better preserve image quality and produce more accurate results. As Fig. 1 shows the comparison RS images with a sampling factor of 4 between the resampled by bicubic and SR reconstructed network, the high-resolution images generated by the SRR network are obviously better and clearer. Therefore, we believe that the SRR network can effectively solve the problems of difficult training and poor segmentation caused by the low resolution of RS images themselves and the complex blurring of edge textures.

In this article, we propose a multitask and multistage joint training strategy (TS) based on SRR task for SS (SRSSTS) of RS images. This TS improves the multiscale and local feature extraction ability of the depth network model, and enables the network to train the same region more effectively under the multiresolution RS image with richer and more delicate texture, so as to solve the problems of blurred edges and inaccurate recognition of local features caused by unclear texture and edge of RS images. The key contributions of the article are as follows.

1) We designed a multistage SS TS called SRSSTS, which utilizes SRR. Our proposed approach, SRSSTS, can be used with all SS backbone networks, and it consistently outperforms single-stage single-task models, providing superior segmentation results. By pre-training the super resolution (SR) reconstruction network and employing multistage progressive joint training of RS images at different resolutions, we obtain high-resolution segmentation results of low-resolution images using joint losses. This approach offers better performance and more accurate segmentation results compared to single-stage and single-task models.
2) We developed a novel GAN-based SRR network that is dominated by perceptual loss. Our generator is an encoder-synthesis network, which focuses on enhancing texture detail features while minimizing smoothing and artifacts in the reconstructed image. To ensure pixel-by-pixel authenticity discrimination, our discriminator uses a U-Net network combined with spectral normalization (SN) and patch discrimination. These advancements have resulted in improved image quality and overall performance of the network. We demonstrated the effectiveness of our proposed SRR network after extensive comparison experiments on five different resolution RS datasets to obtain the best perceptual value learned perceptual image patch similarity (LPIPS).

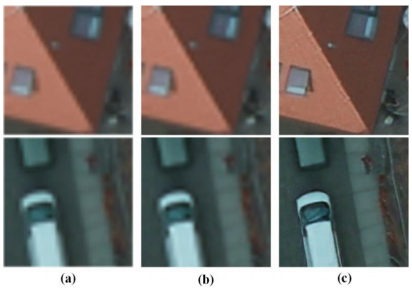


Fig. 1. SRR. (a) Low-resolution RS image. (b) High-resolution image by bicubic resample with a scaling factor of 4. (c) High-resolution image by style-based GAN (ours) with a scaling factor of 4.

3) We successfully demonstrated the efficacy of our proposed SR SS technique (SRSSTS) in improving the performance of RS SS. Our study involves extensive experiments on five RS datasets with different resolutions, replacement experiments using various SS backbone networks, SRR networks, and comparison experiments with other state-of-the-art SR-based segmentation methods. As a result, our SRSSTS-Seg yields the best accuracy and visual appeal compared to other methods.

II. Related Work

This work is related to DL-based multitask and multistage training segmentation methods, DL-based RS single-image SR and SR-based segmentation for RS images.

A. DL-Based Multitask and Multistage Learning

Multitask learning (MTL) is a promising field in machine learning, where domain knowledge is used to improve generalization performance by using supervised signals from the relevant tasks [25]. In MTL, information among tasks is shared, and knowledge is transferred between different tasks. Also, the generalization performance of different tasks is improved due to the sharing of datasets and basic features. Multilabel learning [26] is a form of MTL. MTL also includes multioutput learning, auxiliary training [27], and so on. Sun et al. [28] obtained more accurate segmentation results without additional annotation workload, especially at the object boundaries by MTL with channel-weighted multiscale feature (CMF) module and boundary attention module (BAM). Cheng et al. [29] proposed a multitask learning method of road segmentation, direction estimation, and road edge learning to make our model connect roads reasonably.

Multistage learning refers to the completion of a task in multiple steps. The previous stage provides auxiliary information support for the next stage, which is conducive to the smooth completion of the overall task. It is often combined with MTL. Zhao et al. [30] proposed multistage fusion module for RS image segmentation, where the module fuses complementary information after calibrating the deviation information by filtering the noise from the multimodal data. We have also studied this direction before, based on the edge detection network and SS network, we proposed a two-stage model for high-resolution RS image segmentation tasks, in which the edge detection results are used to guide the prediction of SS results [8].

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Several studies demonstrate that multitask and multistage learning can effectively improve the SS accuracy of images. This article proposes the use of the SRSSTS, a multitask and multistage learning approach, for reconstructing low-resolution RS images into high-resolution ones. The method involves pre-training an SRR network and then sequentially training a multistage SS network using the high-resolution images, and finally obtains high-resolution segmentation results through low-resolution RS images to improve the segmentation accuracy.

B. DL-Based RS Single-Image SR

Most of the current RSISR algorithms are based on supervised learning which is trained with the HR-LR pairs. In this section, we will discuss the CNN-based and GAN-based RSISR methods.

1) CNN-Based RSISR: The study proposed by Leibel and Körner [31] is one of the first CNN-based work applied to an SRSISR task. In this study, the authors explored the CNN method on Sentinel-2 images and showed that the CNN methods can be successfully applied to RS data (Sentinel-2 images) both in Y channel (which is luma components in YCbCr color space) and multispectral (MS) RS images by fine-tuning the SRCNN with the particular dataset. However, they focused only on the third band of the MS images for the performance observation. Tuna et al. [32] applied the SRCNN and VDSR models incorporating IHS transformation on the satellite images acquired by VHR SPOT6 and 7 and the Pleiades 1A and 1B satellites with a scale factor of 2–4. However, due to the feature complexity of the RS images, there is a real need to propose and develop a specific model for RSISR. Xu et al. [33] proposed a deep memory connected network (DMCN) which combines the image detail with environmental information by using local and global memory connections to produce high-quality images. To reduce the memory, the authors proposed downsampling and upsampling units to conduct an hourglass structure. The experimental results show that good results have been achieved on NWPU-RESISC45, UC Merced dataset, and Gaofen-1 testing images with scales of 2–4. Recently, Ren et al. [34] proposed an enhanced residual convolutional neural network (ERCNN) with a dual-luminance scheme (DLS) for RSISR which enhanced the feature flow module and ability to distinguish learning across feature maps. Deeba et al. [35], proposed a transferred wide residual SRSISR deep neural network model (WRSR) which considerably reduced the memory costs and improved the accuracy with the increase of width and decrease in the residual network depth.

2) GAN-Based RSISR: The GAN framework has been recently considered for the RSISR.

A GAN-based method, transferred GAN (TGAN), removed the batch normalization layers to reduce the computational burden and improve the accuracy [36]. The model was first trained on the DIV2K natural images and fine-tuned with the RS dataset. Jiang et al. [37] proposed an EEGAN architecture, in which the generator of EEGAN consists of two main subnetworks: an ultra-dense subnetwork (UDSN) and an edge-enhancement subnetwork (EESN). The experiments on Kaggle open-source dataset, Jilin-1 video satellite images, and Digi-Teleglobe showed superior performance compared to other SR approaches. Gong et al. [38] proposed an Enlighten-GAN for SR task by designing the enlightened blocks, using the self-supervised hierarchical perceptual loss to ensure the improvement of the performance, and employing the internal inconsistency loss and cropping-and-clipping strategy to avoid the seam lines. More recently, Chen et al. [39] effectively improved the EEGAN with the diagonal connections for better information conversion and gradient optimization by applying a 2-D topology and RRDB blocks. Nevertheless, how to super-resolve a VHR image (e.g., 50 or 30 cm) to generate a higher resolution image is still practical and challenging work. In [40], by applying a residual dense network (RDN) in a generator network, the authors proposed a GAN-based approach which provides high performance on classical (objective) evaluation metrics along with high perceptual quality for RSISR. Liu et al. [41] proposed a cascaded conditional Wasserstein GAN (CCWGAN) by designing the residual dense block to obtain HR RS images. Guo et al. [42] developed a cascade generative adversarial nets (CGAN) by designing an edge enhancement module conserving edge details and suppressing the noise.

Based on our analysis, how to make full use of the abundant information contained in the MS images has been limitedly explored and deserves detailed further research in the future.

C. SR-Based RS Segmentation

The SR task is to generate high-resolution images from low-resolution images, which is a significant help to improve the accuracy of RS SS. Jiang et al. [43] proposed SR segmentation with relation calibrating network, where the low-resolution gradually upsampled to high-resolution first, then output the high-resolution segmentation results by feature calibrating module. Abadal et al. [44] proposed a dual-network model for Sentinel-2 imagery segmentation with two branches: the single image SR branch and the SS SR branch. The first branch reconstructs a high-resolution version of the input images, and the second branch predicts a high-resolution segmentation map with a scaling factor of 2. Zhang et al. [20] designed a collaborative network that simultaneously solves the SR SS and SR image reconstruction. This algorithm can obtain high-resolution SS and SRR results by taking relatively low-resolution images as input when high-resolution data are inconvenient or computing resources are limited.

Currently, SS methods based on SR often use SR and SS networks to train collaboratively. However, this approach has limitations in improving segmentation performance and increases computational costs. In particular, the SS loss is typically given greater weight than the SR loss in the definition of the loss function to ensure improved segmentation accuracy. This can limit the ability of the SR network to achieve optimal performance. Our proposed method addresses these limitations by training the SR network separately to achieve the best possible reconstruction quality for high-resolution

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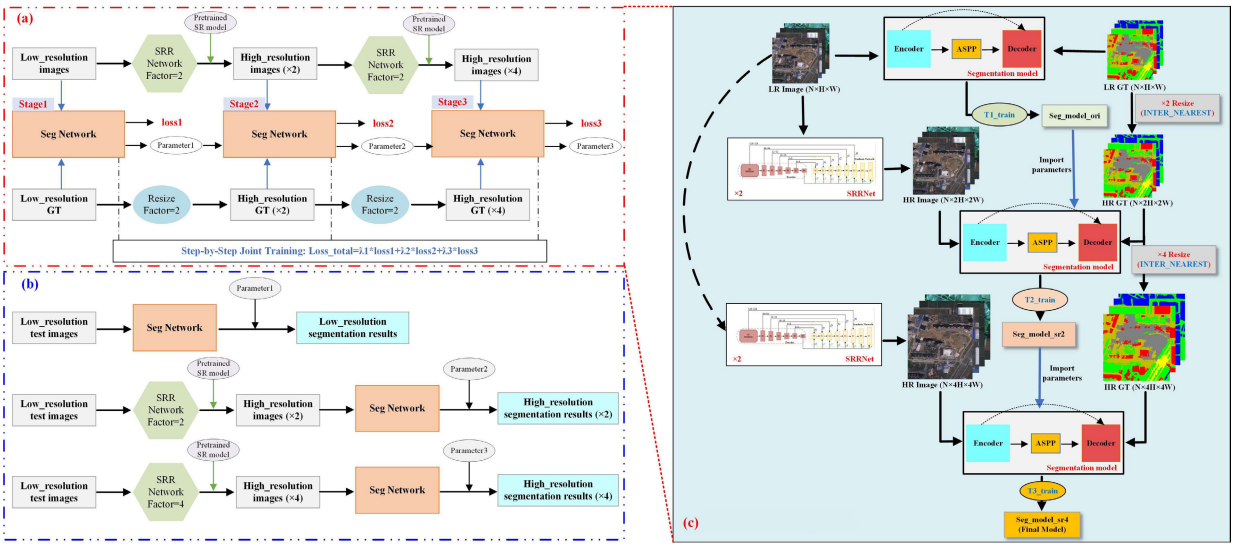


Fig. 2. Training and testing process overview diagram of SRSSTS. (a) Train. (b) Test. (c) Detailed training process.

images. This allows the SS network to learn richer and more accurate features, which can then be used to improve the final segmentation performance. Importantly, our joint-loss SS network does not involve the loss of SR network, which further ensures the quality of the final segmentation results.

III. Methodology

A. Multistage TS Based on SRR

In contrast to the single segmentation training approach, we propose a multistage SRSSTS task of RS images. While the first stage is the same as the previous training model, the last two stages incorporate an SR generation task, i.e., step-by-step optimization and fine-tuning of the model parameters by further feature learning and extraction of the training samples at the larger resolution.

Fig. 2 is a sketch of the training and testing strategy in this article. This is a multistage cyclic progressive TS, in which the model parameters from the previous training stage are transferred to the next high-resolution training stage for optimization and adjustment during each iteration, and each stage generates a loss value and performs a joint loss calculation at the end of each iteration. It should be noted that the SRR network is trained separately in advance to get the pre-trained model parameters, which makes the training process easy by generating clear and accurate high-resolution images at the beginning. This saves both computational effort and training time, and allows the network to focus on the SS task. At the end of training, we will save the segmentation model parameters for each stage and the test images will be segmented at different resolutions according to the requirements. The low-resolution test images are passed through SRRNet with different scale factors to obtain high-resolution images, and then the corresponding network parameters are used to obtain high-precision segmentation results with different resolutions.

B. SRR Network

This article proposes a GAN-based SRR network that consists of two networks: a generative network and a discriminative network. The generative network is designed to create high-resolution images from low-resolution images, while the discriminative network evaluates the authenticity of the generated images to enhance the reconstruction capability of the generative network. The ultimate goal of the network is to produce high-quality images with rich texture features that can enhance image perception.

1) Generative Network: The latent encoding is capable of expressing object features in depth, i.e., it can use less information to express the essence of the object. Therefore, we believe that the effective recovery of latent encoding can improve the ability of the generator to generate images with more realistic and richer texture features. To explore this possibility, we have designed an encoder–decoder generator specifically for the purpose of reconstructing low-resolution images. The encoder is used to extract latent vectors and multiresolution convolutional features from the LR image, which capture important high-level cues and spatial structure. The decoder is responsible for efficient recovery of latent codes and fusion with multiscale features to gradually generate high-resolution features. Fig. 3 illustrates the generative network utilized in this article, which comprises an encoding network and a decoding network (synthesis network).

a) Encoder: The encoding network is responsible for extracting and compressing low-resolution image features, which are subsequently transmitted to the synthesis network to learn and generate high-resolution images. The encoding network is comprised of six blocks (Ei), with $E0$ being an RRDBNet [18] that extracts feature $f0$ from low-resolution images. RRDB consists of multiple densely connected blocks (DB), each DB consists of five convolutional layers and five ReLU activation layers connected by dense jumps, which facilitates the extraction of richer feature details. The resolution of the features is then gradually reduced by

$$fi = Ei(f(i-1)), \quad i \in \{1, 2, \ldots, N\} \tag{1}$$

where Ei, $i \in \{1, 2, \ldots, N\}$, denotes a stack of a stride-2 convolution and a stride-1 convolution. Finally, $E(N+1)$ is used to generate the latent vectors C, which consists a convolution and a fully connected layer. The latent vectors

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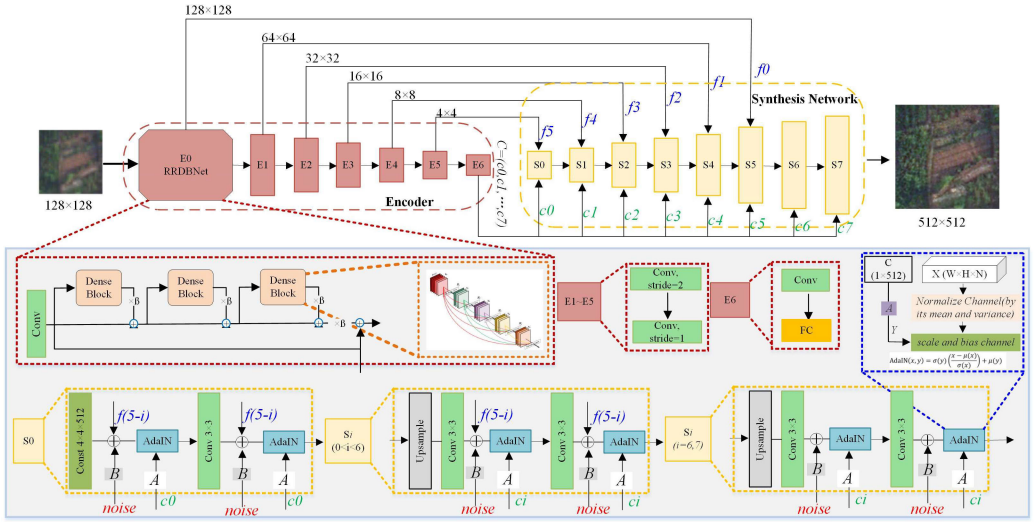


Fig. 3. Structure of generative network in this article. In addition to the latent vectors *ci*, the generator is also conditioned on the multiresolution features *fi*. This example corresponds to an input size of 128 × 128 and an output size of 512 × 512 (4).

in C capture a compressed representation of the images, providing the generative synthesis network with high-level information, where C is a matrix whose columns represent the latent vectors for the synthesis network. To further capture the local structures of the LR image and to provide additional guidance for structure restoration, we also feed multiresolution convolutional features $\{fi\}$ into the synthesis network.

b) Synthesis network: A progressive resolution enhancement generation network that not only enhances the resolution but also allows precise tuning and editing of the properties of the generated images. Instead of taking one single latent vector as the input, each block of the generator takes a different latent vector to improve expressiveness. More specifically, we have $C = (c0, \ldots, ck-1)$ for k blocks, where each ci corresponds to one latent vector, input to *Sk*. We find that this modification leads to outputs with fewer artifacts. This modification is also seen in previous works [45], [46]. In addition to this, we add features *fi* of different scales (resolutions) in the first few blocks for guided generation, instead of noisy random generation. In the generated block S, the first block $S0$ does not contain an upsampling layer, while each subsequent block contains one. Additionally, each block *Sk* includes two AdaIN stylized modules [47]. As shown in (2), with AdaIN capable of rapid generation and supporting detailed attributes such as content color modulation. For the input, A refers to the affine transform obtained by applying a transformation to the latent vector ci, which controls the style of the generated image. On the other hand, B refers to the random noise that has been transformed using a noise transformation, which serves to enhance the details of the generated image

$$\mathrm{AdaIN}(x, y) = \sigma(y)\left(\frac{x - \mu(x)}{\sigma(y)}\right) + \mu(y). \tag{2}$$

σ represents the mean, μ represents the variance, x and y represent the input feature map F and potential vector C, respectively. Realized the transfer of styles (Y to X).

2) Discriminative Network: In order to highlight the high-frequency information (texture details) of the reconstructed HR RS images, this article draws on the idea

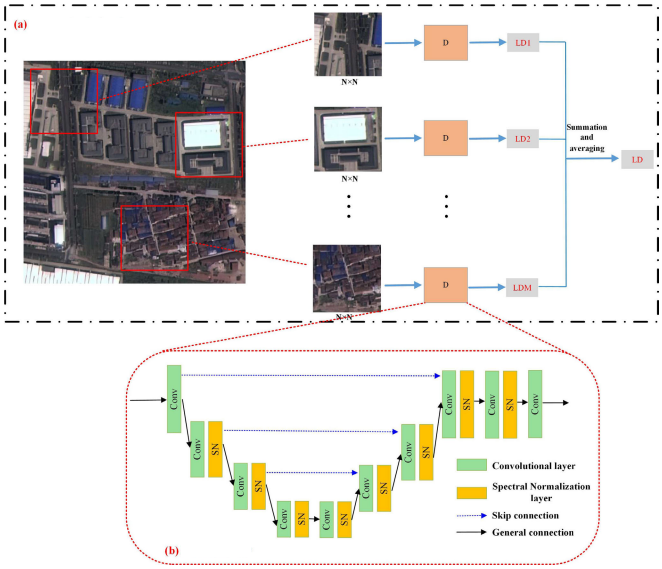


Fig. 4. Structure diagram of patch-based discriminative network in this article. (a) Path-based discriminator, N denotes the size of each patch, M denotes the number of co-patches, and LD denotes the loss value of the discriminator output. (b) Discriminator network: U-Net with SN.

of PatchGAN [48] to divide the images in the discriminator into multiple patches (the number is M) to determine the veracity of each patch ($N \times N, N = 70$), and the veracity responses of all patches are averaged as the final output of the discriminator D. The process of the patch-based discriminator is shown in Fig. 4(a). According to [19], we use U-Net structure to replace VGG structure with more accurate gradient for local textures in discriminator network [see Fig. 4(b)]. The U-Net outputs realness values for each pixel, and can provide detailed per-pixel feedback to the generator. In addition, to improve the training stability, we add SN to U-Net to keep the network parameters within a range so as to stabilize the discriminator training.

C. Loss Function

1) SRR Loss: Following previous works [18], [49], the objective function for StyleGAN-based SRR network consists

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of three terms: L1 loss, perceptual loss [50], and adversarial loss [51]

$$l1 = E_{xi}\left\|\dot{y} - y\right\|_1 \tag{3}$$

$$l_{\text{percep}} = \frac{1}{N}\left\|f\left(\dot{y}\right) - f(y)\right\|_2^2 \tag{4}$$

$$l_{\text{gen}} = \log(1 - D(\dot{y})) \tag{5}$$

$$l_d = \log\left(1 - D\left(\dot{y}\right)\right) + \log D(y) \tag{6}$$

where $l1$ is the content loss that evaluate the 1-norm distance between recovered image $\dot{y}$ and the ground-truth y, l_{percep} is the perceptual loss, l_{gen} is the adversarial loss, l_d is the discriminator loss, $f(\cdot)$ denotes the feature embedding space of the VGG19 [50] network, and D corresponds to the discriminator in this article. So, the resulting objective function is a weighted mean of the three losses

$$l_g = l1 + \alpha l_{\text{percep}} + \beta l_{\text{gen}}. \tag{7}$$

α, β are the coefficients to balance different loss terms, according to [52], we set $\alpha = \beta = 0.2$.

2) Segmentation Loss: The training process generates segmentation losses (cross entropy) at three different resolutions, see Fig. 2. We design the joint loss calculation to optimize the model at different resolutions, and on this premise, we also design adaptive weight coefficients (λ_i) to ensure the accuracy of high-resolution feature segmentation

$$l_{\text{seg}} = \lambda_1 \text{loss1} + \lambda_2 \text{loss2} + \lambda_3 \text{loss3} \tag{8}$$

$$\lambda_1 = 1 - \frac{\text{loss1}}{\text{loss1} + \text{loss2} + \text{loss3}} \tag{9}$$

$$\lambda_2 = 1 - \frac{\text{loss2}}{\text{loss1} + \text{loss2} + \text{loss3}} \tag{10}$$

$$\lambda_3 = 1 - \frac{\text{loss3}}{\text{loss1} + \text{loss2} + \text{loss3}}. \tag{11}$$

IV. Experiment and Results

In this section, we describe in detail the datasets used in our experiment, introduce the evaluation indicators, and present the experimental results in SRR and SRSSTS segmentation.

A. Experimental Data

1) ISPRS Potsdam Dataset (0.05 m): Potsdam dataset we used provides 38 images with three bands (NIR, R, G) from a typical historic city with giant building blocks, narrow streets, and dense settlement structures in Germany with a spatial resolution of 0.05 m. The ground truth for SS includes six common land cover categories: impervious surfaces, low vegetation, trees, buildings, cars, and backgrounds. For training and testing purposes, we cropped each image to dimensions of 128 × 128 and 256 × 256. Specifically, we used 53040, 7456, and 5808 images with 128 × 128 dimensions for training, validation, and testing, respectively. For images with 256 × 256 dimensions, we used 13 260, 1864, and 1452 images for training, validation, and testing, respectively.

2) ISPRS Vaihingen Dataset (0.09 m): Vaihingen dataset comprises 33 images captured in a small village in Germany. Each image includes three bands (NIR, R, G), with a spatial resolution of 0.09 m. The images feature numerous detached buildings and small multistory structures. For training and testing purposes, we cropped each image to dimensions of 128 × 128 and 256 × 256. Specifically, we used 4626, 728, and 880 images with 128 × 128 dimensions for training, validation, and testing, respectively. For images with 256 × 256 dimensions, we used 1156, 182, and 220 images for training, validation, and testing, respectively.

3) LoveDA Dataset (0.3 m): LoveDA is an urban–rural adaptive surface coverage dataset, which is constructed from 0.3 m spatial resolution images and collected in Nanjing, Changzhou, and Wuhan. The data is from Google Earth and was collected in July 2016. The dataset contains a total of seven categories: low vegetation, building, road, bare soil, water, forest, and crops.

1) *Rural:* The training set has 1366 images, the validation set has 992 images, and the test set has 976 images, each with a size of 1024 × 1024. Since the test data has no ground truth to do accuracy evaluation, we randomly select 300 images from the validation dataset as the test set. For training and testing purposes, we cropped each image to dimensions of 128 × 128 and 256 × 256. Specifically, we used 80 760, 44 288, and 19 200 images with 128 × 128 dimensions for training, validation, and testing, respectively. For images with 256 × 256 dimensions, we used 1156, 11 072, and 4800 images for training, validation, and testing, respectively.
2) *Urban:* We used 71 224, 24 128, and 19 200 images with 128 × 128 dimensions for training, validation, and testing, respectively. For images with 256 × 256 dimensions, we used 17 806, 6032, and 4800 images for training, validation, and testing, respectively.

4) WHDLD Dataset (2 m): WHDLD contains 4940 RGB images in 256 × 256 captured by Gaofen-1 Satellite and ZY-3 Satellite over Wuhan urban area. By image fusion and resampling, the resolution of the images is to reach 2 m/pixel. The images are labeled with six classes, i.e., bare soil, building, pavement, vegetation, road, and water. For training and testing purposes, we cropped each image to dimensions of 128 × 128 and 256 × 256. Specifically, we used 17 360, 1600, and 800 images with 128 × 128 dimensions for training, validation, and testing, respectively. For images with 256 × 256 dimensions, we used 4340, 400, and 200 images for training, validation, and testing, respectively.

B. Evaluation Metrics

In segmentation, we use overall accuracy (OA), Recall, F1-score, and mean intersection over union (MIoU) to quantitatively evaluate the results. The accuracy evaluation is usually divided into four categories, which are the number of correctly judged cloud elements (true positive, TP), the number of correctly judged non-cloud elements (true negative, TN), the number of incorrectly judged cloud elements (false positive, FP), and the number of incorrectly judged non-cloud elements (false negative, FN). The four categories form the confusion

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matrix of cloud detection results, and the five accuracy evaluation indexes are calculated according to the confusion matrix, and the specific calculation formulae are shown in following formulas:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{12}$$

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{13}$$

$$P = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \quad F1 = 2 \times \frac{P \times \mathrm{Recall}}{P+\mathrm{Recall}} \tag{14}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}}, \quad \mathrm{MIoU} = \frac{\sum_i^n \mathrm{IoU}}{n}. \tag{15}$$

In SRR, we use LPIPS [53] to evaluate the image quality, while also including PSNR and SSIM indices as references. LPIPS is a method that aims to improve the perceptual quality of generated images by forcing the generator to learn the reverse mapping of the generated image to the ground truth. This approach prioritizes the perceptual similarity between the generated and ground-truth images and has been found to be more consistent with human perception than traditional methods such as PSNR and SSIM. A lower LPIPS value indicates a higher similarity between the two images, while a higher value indicates a greater difference. Overall, LPIPS is an effective tool for evaluating the perceptual quality of generated images. The metric focuses on calculating the distance $d(x, x_0)$ of the activation features after two inputs (x, x_0) are fed into the network (VGG in this article). Specific definitions can be found in [53]

$$d(x, x_0) = \sum_l \frac{1}{H_l W_l} \sum_{h,w} \left\| w_l \bigodot \left(\hat{y}^l_{hw} - \hat{y}^l_{0hw}\right) \right\|_2^2 \tag{16}$$

where $\hat{y}^l, \hat{y}^l_0 \in \mathbb{R}^{H_l \times W_l \times C_l}$ for layer l of VGG. Note that using $w_l = 1 \forall l$ is equivalent to computing cosine distance.

C. Performance on SR-Reconstruction (Ours)

We conducted SRR experiments using five RS datasets: ISPRS Potsdam, ISPRS Vaihingen, WHDLD, LoveDA Rural, and LoveDA Urban. The aim of these experiments was to showcase the effectiveness of the SRR method proposed in this article. We also performed a comparative analysis with several state-of-the-art models such as SRCNN [11], EDSR [16], TTSR [54], ESRGAN [19], SwinIR [55], RBAN-Unet [56], and HAUNet [57]. As RS images are more complex than natural images, we gathered and curated an SR reconstructed RS dataset, which includes NWPU-RESISC45, CHN6, Satellite dataset II (East Asia), ISPRS, and ICCV_data_public. This dataset comprises a total of 10 000 images, where low-resolution images are obtained from high-resolution images by bicubic interpolation. We then performed secondary training on the RS dataset using pre-trained models of each model on natural images. This secondary training allowed us to obtain optimal parameters for each model to achieve the best results for SRR of RS images.

We performed experiments using sampling factors of $\times 2$, $\times 4$, and $\times 8$ to evaluate the effectiveness of our proposed method for SRR of RS images. Both qualitative and quantitative evaluations were conducted, and the results are presented in Figs. 5–8 for the four datasets (Potsdam, Vaihingen, WHDLD, and LoveDA-Rural) under different methods. Our method outperformed the other methods and produced the best visual effect, with results that are closer to the original high-resolution images after resolution enhancement. Our method produces image textures that are more realistic and richer in detail compared to PSNR-oriented models. The edges in our images are clearer, which significantly reduces the smoothing and blurring of the image. Additionally, when compared to ESRGAN, our generated images exhibit less distortion and pseudo-shadow, as our approach avoids overemphasizing noise-rich textures. Overall, our method strikes a good balance between texture convergence and noise suppression, resulting in images that are visually appealing and true to the original high-resolution images.

As we concentrate on generating RS image textures with superior perceptual quality, the quantitative evaluation in this article primarily focuses on the perceptual index, we use LPIPS to evaluate the image quality, while also including PSNR and SSIM indices as references, as depicted in Table I. Our method delivers superior results in perceptual index on five remotely sensed datasets. Compared with the PI-dominated ESRGAN, our method obtains better LPIPS with PSNR enhancement, demonstrating the ability of our method to enhance the resolution on RS images. While SwinIR and two RS SR networks introduced within the past two years have demonstrated impressive SSIM and PSNR scores, they exhibit subpar performance in terms of LPIPS and continue to suffer from the issue of image blurring due to excessive smoothing. In contrast, our SR approach adeptly addresses this challenge, resulting in noticeably improved realism and enhanced texture clarity within the reconstructed images.

D. Performance on SRSSTS

In this section, we present experiments and analysis on multiple aspects to demonstrate the effectiveness and stability of our proposed SRR-based TS for RS SS. These aspects include: 1) conducting segmentation experiments using our TS with advanced SS methods; 2) performing segmentation experiments using various SRR models during the training process; 3) comparing our proposed method with other SS methods that are based on SRR tasks; and 4) conducting ablation experiments on different training stages (with different sampling factors) within our proposed TS.

1) Analysis of SRSSTS With Different Segmentation Methods: To showcase the efficiency and portability of our proposed TS, we applied it to multiple state-of-the-art SS networks, including Deeplabv3+ [58], OCRNet [59], and SegFormer [60]. We further divided the input datasets into 128×128 and 256×256 size, respectively, and performed multiple sets of extensive experiments on five different spatial-resolution RS datasets to validate the effectiveness of our method.

Figs. 9–11 depict the segmentation results of our proposed TS on five RS datasets, using different SS networks. Each

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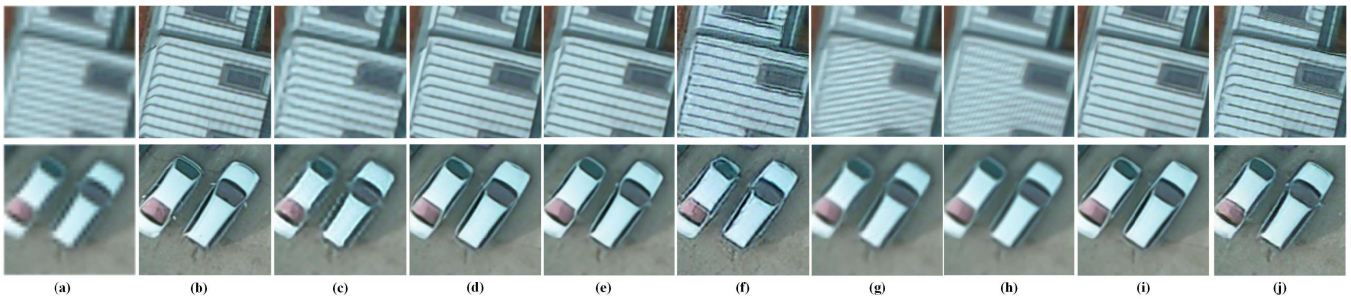


Fig. 5. SRR images generated by different state-of-art methods on the ISPRS Potsdam dataset with a sampling factor of 4. (a) LR image. (b) HR image. (c) SRCNN. (d) EDSR. (e) TTSR. (f) ESRGAN. (g) SwinIR. (h) RBAN-Unet. (i) HAUNet. (j) Ours.

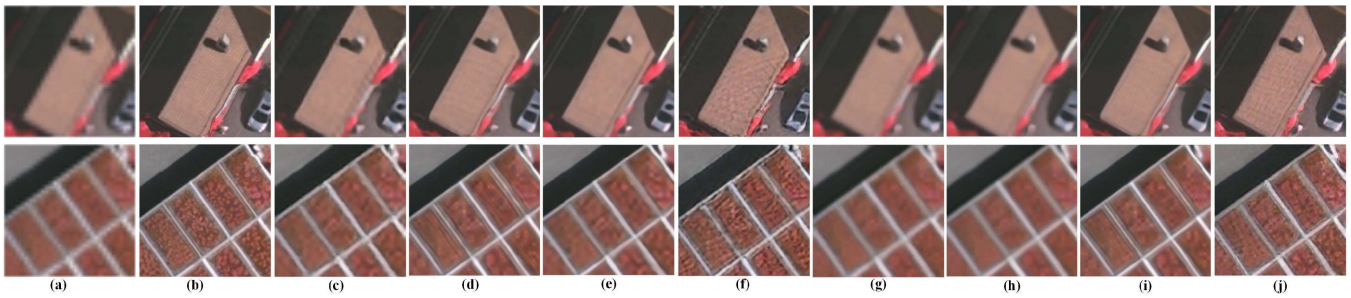


Fig. 6. SRR images generated by different state-of-art methods on the ISPRS Vaihingen dataset with a sampling factor of 4. (a) LR image. (b) HR image. (c) SRCNN. (d) EDSR. (e) TTSR. (f) ESRGAN. (g) SwinIR. (h) RBAN-Unet. (i) HAUNet. (j) Ours.

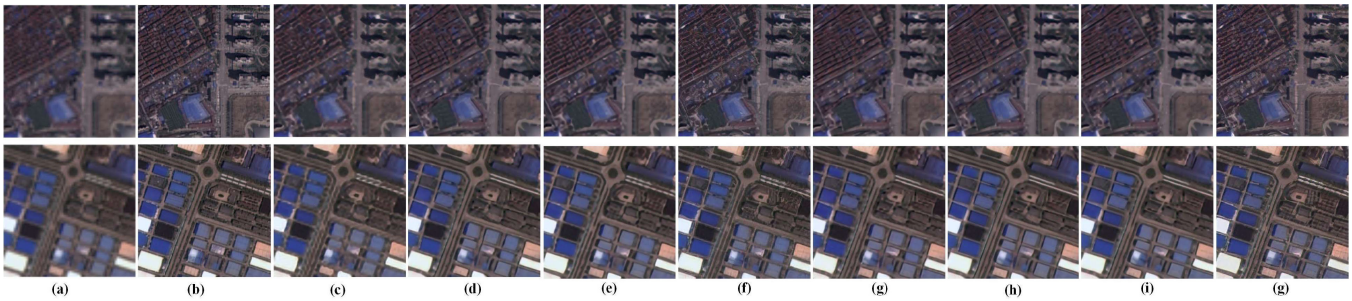


Fig. 7. SRR images generated by different state-of-art methods on the WHDLD dataset with a sampling factor of 4. (a) LR image. (b) HR image. (c) SRCNN. (d) EDSR. (e) TTSR. (f) ESRGAN. (g) SwinIR. (h) RBAN-Unet. (i) HAUNet. (j) Ours.

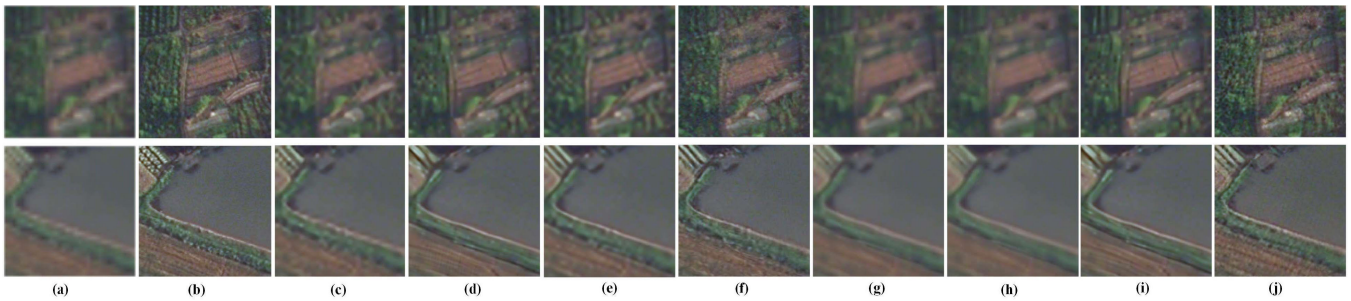


Fig. 8. SRR images generated by different state-of-art methods on the LoveDA Rural dataset with a sampling factor of 4. (a) LR image. (b) HR image. (c) SRCNN. (d) EDSR. (e) TTSR. (f) ESRGAN. (g) SwinIR. (h) RBAN-Unet. (i) HAUNet. (j) Ours.

figure includes two columns with differently colored box lines, representing the SS results of the same network using the normal TS and our proposed SRR TS, respectively. The results demonstrate that the visual segmentation performance of the networks trained with our SRR strategy is significantly superior to the original networks for all datasets. Notably, for low-resolution images, the feature differences between many objects are indistinguishable, but after SRR, these differences become more prominent, thereby enhancing the edge features and improving the network's inter-class feature extraction ability. As a result, the segmentation performance is greatly improved.

Table II presents the quantitative evaluation of our method. We fed two low-resolution images, sized at 128 × 128 px and 256 × 256 px, respectively, into the system, and the output yielded high-resolution results of 512 × 512 px and 1024 × 1024 px, correspondingly. Our method achieves the best performance for OA, Recall, F1, and MIoU regardless of the input size of the low-resolution RS images. Specifically, our method improves MIoU by 2%–4% for three advanced SS networks on the ISPRS and WHDLD datasets, and 4%–7% for the LoveDA dataset. These results demonstrate the effectiveness of the SRR-based TS proposed in this article for improving the performance of SS networks.

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TABLE I

VALIDITY OF SR RESULTS BY DIFFERENT SRR METHODS WITH A SAMPLING FACTOR OF 2, 4, 8 FROM POTSDAM, VAIHINGEN, WHDLD, LOVEDA RURAL AND URBAN DATASETS

methods	Data Set	×2		×4		×8	
		SSIM/PSNR	**LPIPS**	SSIM/PSNR	**LPIPS**	SSIM/PSNR	**LPIPS**
Bicubic	ISPRS Potsdam	0.8636/31.76	0.2965	0.8023/29.26	0.3549	0.7161/24.15	0.3875
SRCNN		0.8758/32.36	0.2688	0.8095/30.15	0.2921	0.7236/24.83	0.3727
EDSR		0.8829/33.95	0.2298	0.8491/31.95	0.2473	0.7359/26.08	0.3286
TTSR		0.8936/34.22	0.2342	0.8534/32.23	0.2503	0.7402/26.27	0.3377
ESRGAN		0.8569/32.79	0.1923	0.7881/29.84	0.2315	0.6933/23.27	0.2966
SwinIR		0.8879/34.37	0.2634	0.8351/32.13	0.3092	0.7396/26.66	0.3688
RBAN		0.9023/34.66	0.2599	0.8342/32.10	0.3064	0.7444/26.75	0.3665
HAUNet		0.9112/35.01	0.2233	0.8572/33.25	0.2465	0.7522/27.11	0.3275
Ours		0.8601/33.06	**0.1877**	0.7898/30.35	**0.2198**	0.6997/23.98	**0.2787**
Bicubic	ISPRS Vaihingen	0.8830/30.77	0.2772	0.8195/27.81	0.3407	0.6329/22.36	0.3775
SRCNN		0.8878/30.69	0.2511	0.8294/28.39	0.2626	0.6247/21.97	0.3469
EDSR		0.9036/31.79	0.2238	0.8560/29.34	0.2461	0.6578/24.59	0.3118
TTSR		0.9046/32.02	0.2277	0.8572/29.41	0.2534	0.6567/24.66	0.3157
ESRGAN		0.8621/29.21	0.1997	0.7827/28.43	0.2329	0.5937/20.67	0.2915
SwinIR		0.8998/32.31	0.2622	0.8497/29.83	0.2917	0.6601/24.70	0.3577
RBAN		0.8926/32.51	0.2594	0.8558/29.78	0.2904	0.6592/24.92	0.3526
HAUNet		0.9055/32.80	0.2223	0.8601/30.00	0.2437	0.6801/25.21	0.3125
Ours		0.8627/29.07	**0.1932**	0.8096/28.57	**0.2234**	0.6041/21.02	**0.2866**
Bicubic	WHDLD	0.8251/31.89	0.3247	0.7577/29.36	0.3769	0.5698/21.35	0.4638
SRCNN		0.8324/32.04	0.3056	0.7541/29.28	0.3436	0.5732/21.48	0.4122
EDSR		0.8567/32.57	0.2832	0.7872/29.96	0.3204	0.5967/22.31	0.3854
TTSR		0.8592/32.61	0.2879	0.7892/30.03	0.3317	0.6014/22.57	0.3927
ESRGAN		0.8031/31.67	0.2648	0.7207/29.07	0.3101	0.5635/21.13	0.3543
SwinIR		0.8498/33.15	0.2861	0.7799/30.42	0.3272	0.5927/22.98	0.3821
RBAN		0.8536/33.20	0.2902	0.7826/29.97	0.3326	0.6022/22.66	0.3877
HAUNet		0.8552/33.39	0.2845	0.7844/29.98	0.3319	0.6069/22.79	0.3752
Ours		0.8127/32.01	**0.2526**	0.7283/29.16	**0.2973**	0.5831/21.97	**0.3305**
Bicubic	LoveDA Rural	0.7267/28.68	0.2715	0.6449/27.67	0.3810	0.5859/22.38	0.4222
SRCNN		0.7158/28.61	0.2672	0.6280/26.90	0.3460	0.5927/22.67	0.4028
EDSR		0.7369/29.14	0.2445	0.6881/28.00	0.3168	0.6015/23.21	0.3724
TTSR		0.7377/29.23	0.2501	0.6912/28.09	0.3223	0.6034/23.27	0.3856
ESRGAN		0.7098/28.78	0.2178	0.5964/27.58	0.2812	0.5757/20.33	0.3368
SwinIR		0.7402/29.68	0.2689	0.6957/28.52	0.3462	0.6047/23.74	0.3956
RBAN		0.7441/29.57	0.2693	0.7160/28.47	0.3556	0.6156/23.82	0.3972
HAUNet		0.7512/30.01	0.2432	0.7244/28.88	0.3148	0.6209/24.13	0.3693
Ours		0.7164/28.87	**0.2109**	0.6213/27.78	**0.2661**	0.5910/21.22	**0.3147**
Bicubic	LoveDA Urban	0.8015/29.66	0.2722	0.6987/27.33	0.3903	0.6133/24.88	0.4001
SRCNN		0.7945/28.97	0.2614	0.6825/26.90	0.3299	0.6007/24.35	0.3844
EDSR		0.8337/31.02	0.2394	0.7388/28.61	0.3025	0.6329/26.02	0.3527
TTSR		0.8356/31.77	0.2475	0.7423/28.76	0.3056	0.6318/26.12	0.3645
ESRGAN		0.8044/30.17	0.2078	0.6853/28.36	0.2633	0.6027/25.36	0.3127
SwinIR		0.8401/32.01	0.2658	0.7476/28.92	0.3491	0.6356/26.41	0.3872
RBAN		0.8523/32.35	0.2623	0.7647/28.86	0.3484	0.6436/26.75	0.3769
HAUNet		0.8386/32.47	0.2467	0.7357/28.91	0.3301	0.6378/26.89	0.3547
Ours		0.8036/30.15	**0.2044**	0.6882/28.55	**0.2597**	0.6119/25.40	**0.2956**

Although SegFormer achieves great accuracy improvement on the RS dataset, the segmentation performance of the Deeplabv3+ network with our TS surpasses even that of SegFormer. Moreover, with our SRSSTS, SegFormer still outperforms other SS networks in terms of segmentation performance.

It is important to mention that when evaluating segmentation accuracy, we excluded the sixth noisy background class (clutter) from ISPRS dataset, so that both Potsdam and Vaihingen datasets have higher individual class scores and MIoU above 70%.

Table III presents the intersection over union (IoU) scores for each category (excluding clutter) of the ISPRS Potsdam and Vaihingen datasets, obtained through the Deeplabv3+ and SRSSTS-v3+ methods. The corresponding bar statistics are illustrated in Fig. 12, which clearly demonstrate that our proposed SRSSTS TS yields a more significant improvement in accuracy for each category of the dataset, particularly

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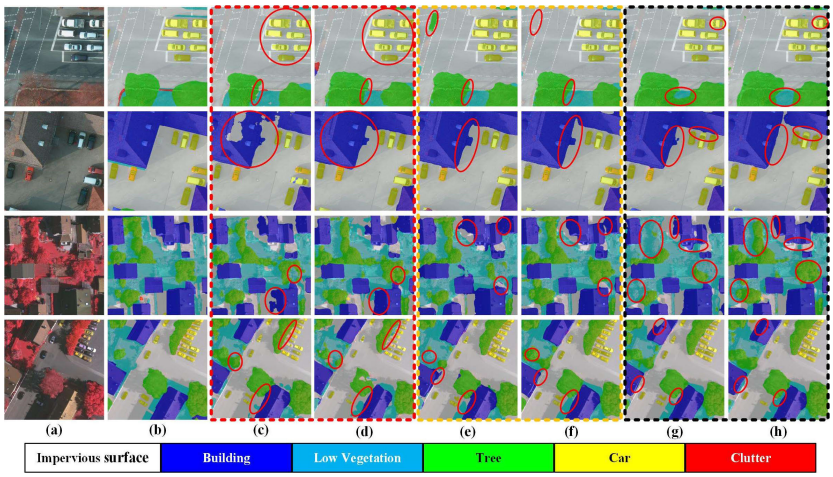


Fig. 9. Visual comparison of SS results (different segmentation networks) with an input resolution of 128 × 128 on the ISPRS Potsdam and Vaihingen test dataset. (a) Input image. (b) Ground truth. (c) Deeplabv3+. (d) SRSSTS-Deeplabv3+. (e) OCRNet. (f) SRSSTS-OCRNet. (g) SegFormer. (h) SRSSTS-SegFormer.

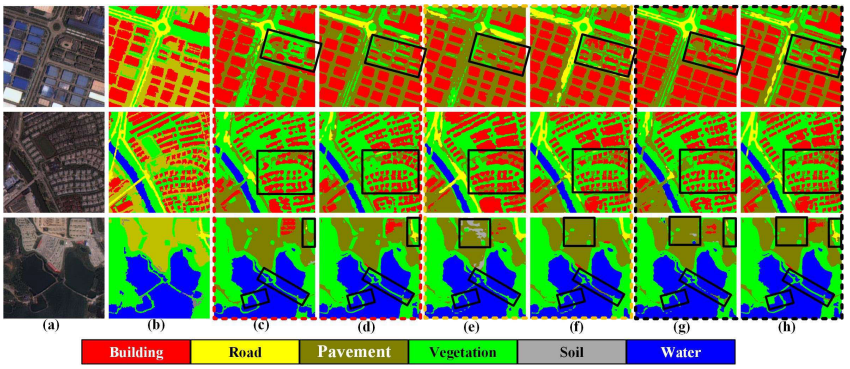


Fig. 10. Visual comparison of SS results (different segmentation networks) with an input resolution of 128 × 128 on the WHDLD test dataset. (a) Input image. (b) Ground truth. (c) Deeplabv3+. (d) SRSSTS-Deeplabv3+. (e) OCRNet. (f) SRSSTS-OCRNet. (g) SegFormer. (h) SRSSTS-SegFormer.

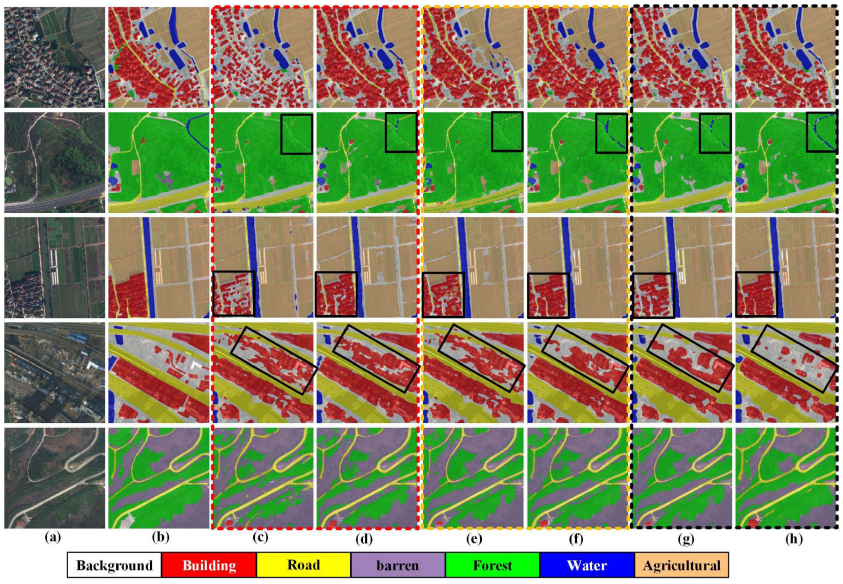


Fig. 11. Visual comparison of SS results (different segmentation networks) with an input resolution of 128 × 128 on the LoveDA Rural and Urban test dataset. (a) Input image. (b) Ground truth. (c) Deeplabv3+. (d) SRSSTS-Deeplabv3+. (e) OCRNet. (f) SRSSTS-OCRNet. (g) SegFormer. (h) SRSSTS-SegFormer.

for the car category. The IoU improvement for car category is as high as 16.71% and 8.87% for the two datasets, respectively, indicating that our SRSSTS approach effectively enhances the precise recognition of small targets in RS images.

2) Analysis of SRSSTS With Different SRR Net: To further showcase the effectiveness of our SRR network, we conducted comparative experiments using various SRR networks (SRCNN, EDSR, TTSR, ESRGAN, SwinIR, RBAN, HAUNet, and Ours) for our proposed SRSSTS. As an example, we used

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TABLE II

VALIDITY OF SRSSTS SEGMENTATION RESULTS BY DIFFERENT SEGMENTATION METHODS WITH 128 × 128 INPUT AND 256 × 256 INPUT FROM POTSDAM, VAIHINGEN, WHDLD, LOVEDA RURAL AND URBAN DATASETS

Methods	**Data Set**	Input size: **128×128**				Input size: **256×256**			
		OA (%) / Recall (%) / F1 (%) / MIoU (%)							
Deeplabv3+ [58]	ISPRS Potsdam	82.94	80.89	82.57	72.32	86.69	86.30	87.47	76.27
Deeplabv3+**(ours)**		**86.35**	**85.91**	**84.28**	**76.50**	**88.86**	**88.57**	**88.69**	**78.44**
OCRNet [59]		85.42	82.36	83.85	73.03	87.97	87.22	86.07	77.03
OCRNet**(ours)**		**87.44**	**86.84**	**85.87**	**76.48**	**89.33**	**88.63**	**89.17**	**79.89**
SegFormer [60]		86.54	85.92	85.61	75.24	89.27	88.54	88.03	78.57
SegFormer**(ours)**		**87.90**	**88.27**	**87.53**	**78.64**	**90.66**	**89.31**	**89.22**	**80.08**
Deeplabv3+ [58]	ISPRS Vaihingen	85.78	80.51	82.19	71.89	86.77	82.81	84.96	74.30
Deeplabv3+**(ours)**		**86.99**	**83.73**	**85.47**	**76.91**	**88.09**	**86.28**	**86.76**	**76.91**
OCRNet [59]		86.26	81.84	83.87	72.09	87.15	82.54	84.33	74.86
OCRNet**(ours)**		**88.38**	**86.67**	**87.30**	**77.71**	**90.07**	**87.44**	**87.92**	**78.56**
SegFormer [60]		87.32	83.83	85.36	74.81	88.27	85.96	87.32	76.51
SegFormer**(ours)**		**88.66**	**86.88**	**87.24**	**78.05**	**90.29**	**87.86**	**88.43**	**79.12**
Deeplabv3+ [58]	WHDLD	80.69	70.22	73.35	60.20	83.17	72.66	75.21	63.04
Deeplabv3+**(ours)**		**83.50**	**75.82**	**76.61**	**64.09**	**85.22**	**76.86**	**78.72**	**65.47**
OCRNet [59]		82.39	73.66	74.52	61.71	83.91	74.27	75.74	63.57
OCRNet**(ours)**		**83.91**	**75.86**	**75.63**	**63.03**	**85.36**	**77.11**	**78.85**	**65.92**
SegFormer [60]		82.92	72.38	73.15	62.86	84.23	74.17	75.55	64.07
SegFormer**(ours)**		**84.73**	**76.08**	**76.97**	**65.58**	**86.01**	**78.26**	**78.83**	**66.33**
Deeplabv3+ [58]	LoveDA Rural	79.75	69.94	75.53	61.87	81.65	72.69	77.43	63.27
Deeplabv3+**(ours)**		**81.91**	**74.24**	**78.03**	**65.09**	**84.57**	**77.22**	**81.15**	**68.28**
OCRNet [59]		80.18	72.51	76.13	62.18	82.03	75.21	78.33	64.40
OCRNet**(ours)**		**84.15**	**75.69**	**80.24**	**67.84**	**86.21**	**79.86**	**82.27**	**69.24**
SegFormer [60]		81.46	72.42	76.77	63.39	82.75	75.31	78.65	65.09
SegFormer**(ours)**		**84.45**	**76.03**	**80.63**	**68.32**	**86.06**	**79.77**	**83.00**	**70.12**
Deeplabv3+ [58]	LoveDA Urban	77.47	64.78	72.16	58.56	81.07	74.28	76.28	62.27
Deeplabv3+**(ours)**		**81.10**	**73.64**	**78.64**	**65.23**	**83.33**	**79.02**	**80.15**	**68.37**
OCRNet [59]		79.87	67.26	75.33	60.14	80.98	73.96	77.39	63.11
OCRNet**(ours)**		**82.24**	**75.32**	**79.01**	**66.07**	**83.14**	**78.27**	**80.66**	**70.28**
SegFormer [60]		80.65	71.76	76.28	63.26	81.26	75.22	78.44	65.39
SegFormer**(ours)**		**82.49**	**77.48**	**81.00**	**68.82**	**83.74**	**80.33**	**83.25**	**71.96**

TABLE III

IOU (%) OF SS RESULTS BY SRSSTS-V3+ COMPARED WITH DEEPLABV3+ IN ISPRS DATASET

Dataset	Method	Surface	Building	Low vegetation	Tree	Car
ISPRS Potsdam	Deeplabv3+	80.86	86.66	**61.77**	67.80	64.50
	SRSSTS-v3+	**80.89**	**89.64**	58.93	**71.85**	**81.21**
ISPRS Vaihingen	Deeplabv3+	79.07	84.72	67.96	64.14	63.57
	SRSSTS-v3+	**81.90**	**88.43**	**71.10**	**70.67**	**72.44**

the Deeplabv3+ network and tested our approach on three different datasets: the ISPRS Potsdam, WHDLD, and LoveDA Urban datasets. These datasets are selected to evaluate the performance of our SRR under the SS task. By comparing the performance of our SRR with other methods on these datasets, we aim to demonstrate the superiority of our approach. It is crucial to carefully analyze and interpret the results to derive meaningful conclusions and provide recommendations for future research.

Table IV displays the results of our comparison, with the Deeplabv3+ network serving as the baseline. The evaluation metrics chosen are F1 and MIoU, and the input image size is set to 128 128 pixels for low-resolution images. Our proposed TS demonstrates an improvement over the baseline approach, irrespective of the SRR network utilized. Moreover, the TS based on our specially designed SRR network yields the highest score among all approaches.

As SRR networks are categorized into PSNR-dominated and perception-dominated, we conducted an investigation into the

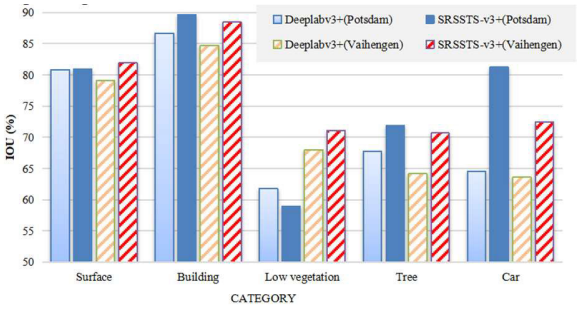


Fig. 12. IoU scoring of each category on the ISPRS Potsdam and Vaihingen test set in Deeplabv3+ and SRSSTS-v3+.

effectiveness of both types in aiding the SS task. We evaluated the segmentation accuracy (MIoU) based on the training strategies of different SRRs and recorded their perception scores (LPIPS), as illustrated in Fig. 13. The results demonstrate that the SRR network contributes significantly to the SS task, as evidenced by the lower LPIPS values. Interestingly, several

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TABLE IV
VALIDITY OF SS RESULTS BY SRSSTS WITH DIFFERENT SRR NETWORKS WITH 128 × 128 INPUT

Methods	Dataset	F1	MIoU	Dataset	F1	MIoU	Dataset	F1	MIoU
baseline		82.57	72.32		73.35	60.20		72.16	58.56
SRSSTS with SRCNN-based		83.11	73.87		73.97	60.98		75.01	61.42
SRSSTS with EDSR-based	**ISPRS**	83.44	74.89	**WHDLD**	74.83	62.27	**LoveDA**	76.69	63.32
SRSSTS with TTSR-based	**Potsdam**	83.36	74.66		74.81	62.05	**Urban**	76.38	63.17
SRSSTS with ESRGAN-based		83.86	75.75		75.66	63.46		78.03	64.51
SRSSTS with SwinIR-based		83.07	73.96		74.29	61.77		74.62	60.96
SRSSTS with RBAN-based		83.19	74.23		74.06	61.58		74.87	61.15
SRSSTS with HAUNet-based		83.67	75.12		74.09	61.63		74.98	61.39
SRSSTS with ours		**84.28**	**76.50**		**76.61**	**64.09**		**78.64**	**65.23**

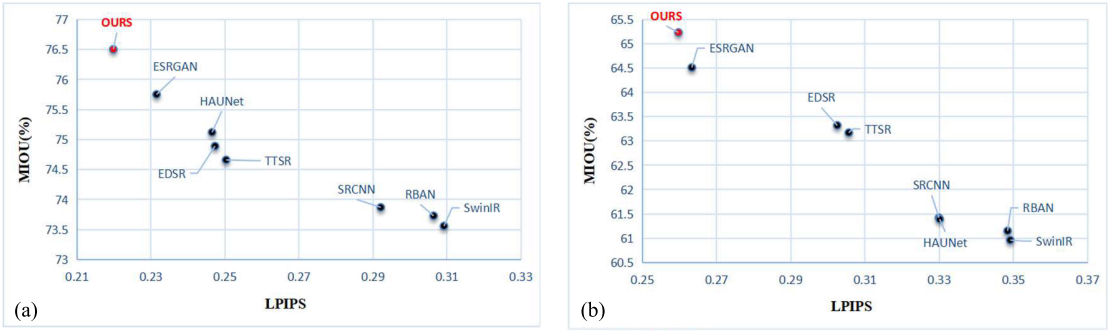


Fig. 13. LPIPS and MIOU statistics for different SRR-based training strategies in two datasets. (a) Potsdam. (b) LoveDA-Urban.

models dominated by PSNR perform poorly in SS, whereas models that prioritize visual perception perform better. Our proposed network achieved the lowest LPIPS score, indicating better visual perception, and its TS has proven effective in improving SS performance.

3) Comparison With Other State-of-the-Art SR-Based Segmentation Methods: We also compare advanced SR-based SS methods in recent years, including RCNet [43], DPN [44], and ColNet [20], where the SS network used in the method proposed in this article is SegFormer. The comparison results are shown in Fig. 14 and Table V. Compared to other networks that incorporate both SR and SS branches, our proposed SRR-based multistage TS outperforms them, demonstrating significant improvements in both F1 and MIoU. We believe that dual-branch multitask learning requires training both the SR and SS networks, which slows down the overall training speed and distracts the network, as it has to focus on both generating SR results and segmentation results simultaneously. On the other hand, our TS involves pre-training the SR network to an optimal state individually. At the beginning of the training, the network receives the best SR result corresponding to the input image, and there is only a segmentation loss. Therefore, the network only needs to focus on producing accurate segmentation results, which enhances its efficiency and effectiveness. In addition, we also conducted efficiency calculations and found that our method training takes the shortest time.

4) Analysis of Different Training Stages in SRSSTS: In this study, we investigated the impact of different stages of SRSSTS on SS enhancement. We conducted experiments using the Deeplabv3+ network on five RS datasets divided into four stages, each with different input and output resolutions. Specifically, stage 1 had an input and output resolution of 128 × 128 (named stage1-x1), stage 2 had an input resolution of 128 × 128 and output resolution of 256 × 256 (named stage2-x2), stage 3 had an input resolution of 128 × 128 and output resolution of 512 × 512 (named stage3-x4), and stage 4 had an input resolution of 128 × 128 and output resolution of 1024 × 1024 (named stage4-x8). The results, presented in Table VI, show that stage 3 achieved the best boosting performance, while the segmentation boosting ability of stage 4 was significantly weaker and even lower than that of stage 2. We believe that the quality of the reconstructed high-resolution image plays a crucial role in the SS ability under this TS. Specifically, when the SR factor reaches 8, the quality of the reconstructed high-resolution image decreases significantly in terms of visual perception and structural relevance. This inferior image quality is much worse than the quality obtained when the SR factor is 2 or 4, which is the main reason for the reduced segmentation ability observed at stage 4. Overall, our study suggests that the SRR-based TS is closely related to the quality of the reconstructed high-resolution image, which, in turn, affects the SS ability.

Based on the analysis presented above, our TS identifies three stages that yield the best segmentation results. These stages are characterized by an SR factor of 2 or 4 and output resolutions of $2H \times 2W$ or $4H \times 4W$. These stages provide a good balance between image quality and segmentation performance, which is crucial for the success of RS applications that rely on accurate SS. Our findings suggest that using an SR factor of 8 may not be the best option when aiming for optimal segmentation performance.

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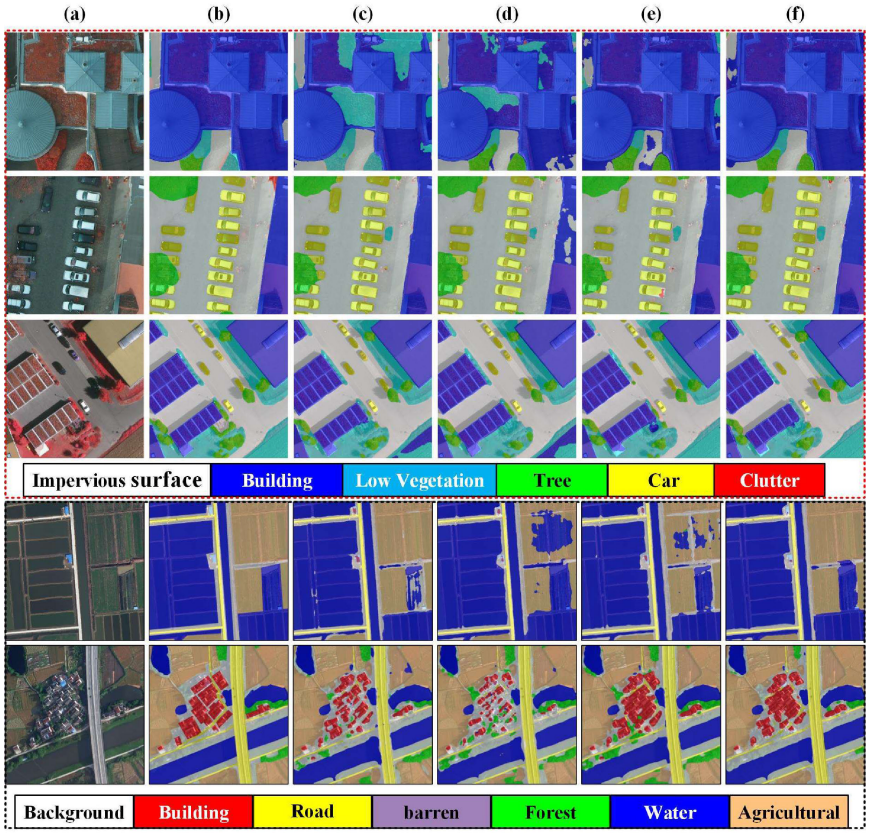


Fig. 14. Visual comparison of different SR-based SS methods with an input resolution of 128 × 128 on the ISPRS (Potsdam, Vaihingen), and LoveDA Rural test dataset. (a) Input image. (b) Ground Truth. (c) Results of RCNet [43]. (d) Results of DPN [44]. (e) Results of ColNet [20]. (f) Results of SRSSTS-SegFormer.

TABLE V

VALIDITY OF SS RESULTS BY DIFFERENT SR-BASED SEGMENTATION METHODS WITH 128 × 128 INPUT (RT: RUN TIME OF TRAINING)

Methods	Dataset	F1	MIoU	Dataset	F1	MIoU	Dataset	F1	MIoU	RT (min)
RCNet [43]		86.18	76.22		86.35	75.66		78.42	64.88	212
DPN [44]	**ISPRS**	85.47	74.96	**ISPRS**	85.27	74.98	**LoveDA**	75.20	62.72	185
ColNet [20]	**Potsdam**	86.51	78.05	**Vaihingen**	87.01	76.97	**Rural**	78.34	66.29	242
SRSSTS-SegFormer		**87.53**	**78.64**		**87.24**	**78.05**		**80.63**	**68.32**	**158**

Note: RT refers to the time (*min*) taken from network training to convergence.

V. DISCUSSION

The SRSSTS proposed in this article can lead to more accurate and stable results obtained by advanced SS networks, which are mainly influenced by the two leading networks in the model, the SRR network and the SS network. The better the learning and expression ability of the two networks, the better the overall segmentation result. Through experiments, we can find that combining different SRR networks and SS networks yields different SS results. In the case where the latter is already mature, this is the reason why we have devoted ourselves to the study of improving the SR reconstruction capability.

At this point, SR networks have become quite mature with two main branches: one that prioritizes PSNR-based reconstruction and another that prioritizes perceptual-loss-based reconstruction. In this study, different types of SR networks were tested and analyzed, and it was discovered that the high-resolution images produced by the perceptually dominated SRR networks exhibit more texture and detail. This more accurately reflects the features of the ground, and can better improve the SS of RS data. Additionally, the experiments in this study demonstrate that the lower the LPIPS value, the stronger the SRSSTS boosting capability. This is the primary motivation for our efforts to enhance the perceptual quality of reconstructed high-resolution images. Moreover, when it comes to selecting SR factors, our analysis shows that factors 2 and 4 are more effective at enhancing RS segmentation, with factor 4 reaching saturation. Expanding the SR factors beyond this point is not conducive to further improvement in segmentation accuracy, due to factors such as distortion and artifacts caused by excessively large SR factors.

Our proposed method is distinct from other SR-based segmentation techniques in that most of them involve simultaneously training the SR and SS networks, resulting in increased computational complexity. In order to address this issue, the SR network needs to be lightweight, but this

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TABLE VI
VALIDITY OF SEGMENTATION RESULTS IN DIFFERENT TRAINING STAGE BASED ON DEEPLABV3+ WITH 128 × 128 INPUT

Dataset	Evaluation Index	Different Training Stage			
		Stage1-x1 (Output:128×128)	Stage2-x2 (Output:256×256)	Stage3-x4 (Output:512×512)	Stage4-x8 (Output:1024×1024)
ISPRS Potsdam	F1 (%) / MIoU (%)	82.57 72.32	83.92 74.77	**84.28** **76.50**	82.89 74.06
ISPRS Vaihingen		82.19 71.14	85.01 73.75	**85.47** **74.97**	83.25 73.29
WHDLD		73.35 60.20	75.11 62.96	**76.61** **64.09**	70.56 59.27
LoveDA Rural		75.53 61.87	77.57 63.88	**78.03** **65.09**	76.21 61.99
LoveDA Urban		72.16 58.56	76.92 62.44	**78.64** **65.23**	74.05 60.12

TABLE VII
CALCULATION EFFICIENCY OF DIFFERENT TRAINING METHODS FOR SRSSTS (WITH POTSDAM DATASET)

Methods	MIoU (%)	RT (min)
SRSSTS-v3+ (tog)	75.67	196
SRSSTS-v3+ (sep)	76.50	142
SRSSTS-Segformer (tog)	78.12	212
SRSSTS-Segformer (sep)	78.64	158

Note: RT refers to the time (*min*) taken from network training to convergence.

compromises the quality of the SR reconstructed images, see Table VII. In contrast, our method trains the SRR network separately from the SS network, and only uses the output of the SRR network as input for the SS network, as shown in Table VII, training SR and SS separately (sep) takes less time and improves accuracy compared to training together (tog). As a result, the SRR network does not need to be quantized lightly, and improving the quality of the reconstructed images can be further prioritized to ensure maximum assistance in SS. After experiments, it has been proved that our strategy has a more obvious enhancement effect on all RS images between centimeter to meter resolution.

VI. CONCLUSION

In this article, we propose SRSSTS for SS of RS. The strategy consists of learning two tasks, including SRR and SS. The reconstructed high-resolution images are used as inputs to assist the SS network in learning multiscale features, and a multistage joint training is proposed to further improve the accuracy of high-resolution segmentation results. SRSSTS is generic and can be applied to any SS backbone network, which is also verified by extensive experiments on five RS datasets (ISPRS Potsdam, Vaihingen, WHDLD, LoveDA Rural and Urban) with different resolutions. Through experimental analysis, our approach has obvious superiority over the single-stage single-task SS model. Based on the advanced SegFormer backbone network, the five datasets are able to obtain IOU scores of 78.64%, 78.05%, 65.58%, 68.32%, and 68.82% with small image training samples of 128 × 128 pixel, respectively. In this work, we also found that different SR models can have a large impact on segmentation results, and the higher the perceptual power, the greater the high-resolution impact helps in SS performance improvement. And our designed StyleGAN SRR network dominated by perceptual loss also achieves the best perceptual score (LPIPS) compared to other networks. In addition, we find that different SR scale factors have different improvement on the performance of RS SS. Notably, the most significant improvement in segmentation performance occurs when the scale factor is gradually increased to 4. However, beyond this point, further increasing the scale factor does not contribute significantly to improving the segmentation performance.

In the future, we will continue to work on multitask learning of SR reconstruction task and SS task, so that the two can be better combined to build a high-efficiency and high-precision segmentation method.

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Baikai Sui received the B.S. and M.S. degrees from the Remote Sensing Science and Technology Department, Shandong University of Science and Technology, Qingdao, China, in 2018 and 2021, respectively. He is currently pursuing the Ph.D. degree in surveying and mapping science and technology with Southwest Jiaotong University, Chengdu, China.

His research interests include digital image processing and deep learning, including convolutional neural network classification, semantic segmentation, and image generation based on adversarial neural network.



Yungang Cao (Member, IEEE) received the M.S. degree in geodesy and survey engineering from Southwest Jiaotong University, Chengdu, China, in 2003, and the Ph.D. degree in cartography and geographic information systems from the Chinese Academy of Sciences, Beijing, China, in 2006.

He is currently an Associate Professor with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University. His research interests include computer vision, deep learning, and remote sensing image processing.



Haibo Cheng received the B.S. degree from the Remote Sensing Science and Technology Department, Shandong University of Science and Technology, Qingdao, China, in 2022. He is currently pursuing the M.S. degree in surveying and mapping science and technology with Southwest Jiaotong University, Chengdu, China.

His research interests include digital image processing and deep learning.



Shuang Zhang received the B.S. degree in surveying and mapping engineering from the Department of Surveying and Mapping Engineering, Xinjiang University, Urumqi, China, in 2021. He is currently pursuing the M.S. degree in resources and environment with Southwest Jiaotong University, Chengdu, China.

His research interests include digital image processing and deep learning.



Jun Zhu received the M.S. degree in geodesy and survey engineering from Southwest Jiaotong University, Chengdu, China, in 2003, and the Ph.D. degree in cartography and geographic information systems from the Chinese Academy of Sciences, Beijing, China, in 2006.

From 2007 to 2008, he was a Post-Doctoral Research Fellow at the Chinese University of Hong Kong, Hong Kong. Currently, he is a Professor with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University. His research interests include computer vision, 3-D GIS technology and virtual geographic environments.



Yakun Xie received the B.S. degree in survey engineering from the School of Survey Engineering, Henan University of Urban Construction, Pingdingshan, China, in 2015, and the M.S. degree in geodesy and survey engineering from the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University, Chengdu, China, in 2018, and the Ph.D. degree in surveying and mapping from Southwest Jiaotong University, in 2022.

He is currently an Assistant Professor with the Faculty of Geosciences and Environmental Engineering, Southwest Jiaotong University. His research interests include remote sensing image processing, deep learning, and computer vision.

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